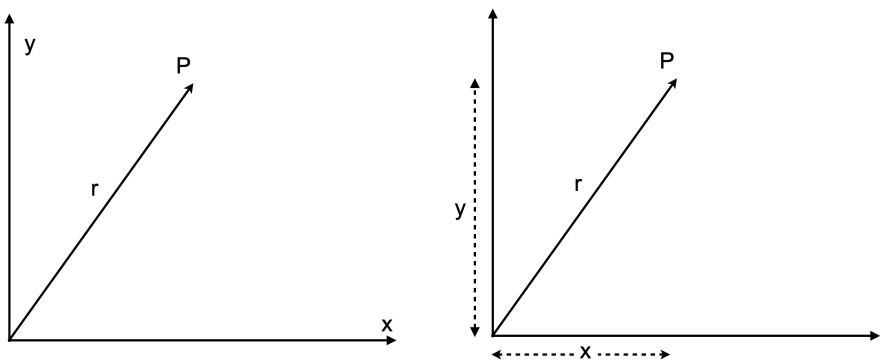
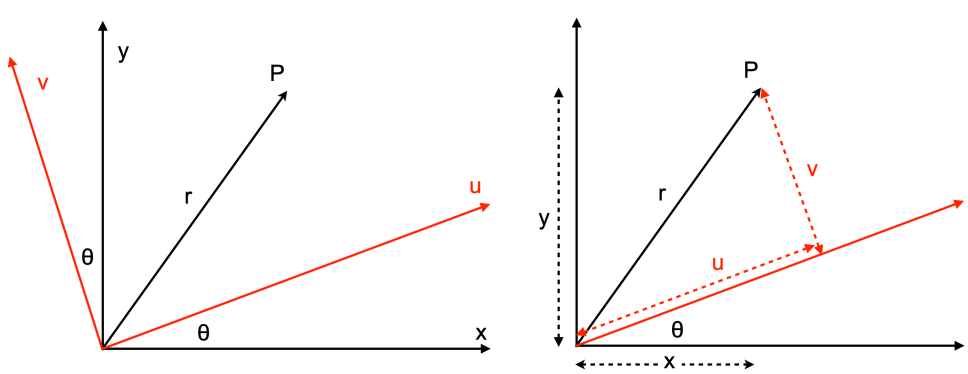
Rotation

Let's begin with the standard picture from analytical geometry, courtesy of Fermat and Descartes.



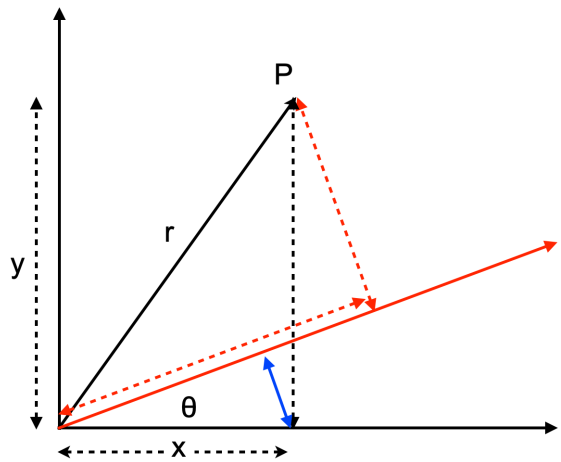
We have a point P that lies a distance r from the origin. We draw orthogonal axes called x and y. Re-using the names, in the xy-coordinate system, $P = (x, y)$.



Now consider some other coordinate system u, v, also with orthogonal

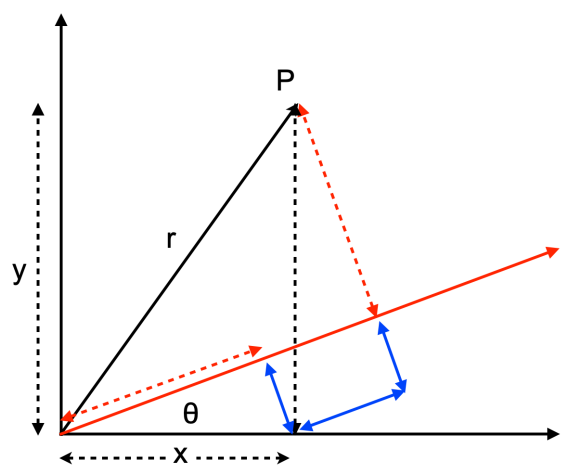
axes but rotated counter-clockwise by an angle θ. In the new system, $P = (u, v)$.

Our goal is to find a formula that relates x and y to u and v.

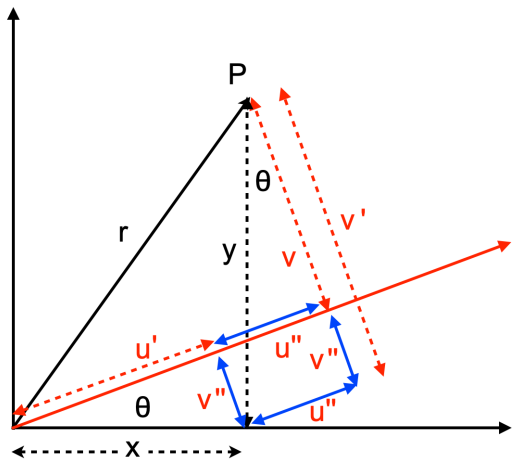


We're going to draw a line that makes x the *hypotenuse* of a right triangle. That's the blue line in the figure above. This is the key step.

Now, just fill in a rectangle.



Finally, put some labels on the segments of lines.



$$u' = x\cos\theta \qquad u'' = y\sin\theta$$

$$v' = y\cos\theta \qquad v'' = x\sin\theta$$

We can write u as the sum of two segments:

$$u = u' + u'' = x\cos\theta + y\sin\theta$$

We write v as the difference of two segments.

$$v = v' - v'' = y\cos\theta - x\sin\theta$$

A convenient notation is to write this as a matrix that multiplies a vector:

$$\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$

This is the matrix for counter-clockwise rotation of coordinates, R_{ccw}

However, alternatively, this can be viewed as a clockwise rotation of the point P.

How about rotation in the other direction?

If you know about matrices, you will realize that the two matrices must be inverses. To invert a 2×2 matrix, transpose it and divide by the determinant (which is 1 here). So

$$R_{cw} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

We can also check to make sure that these work correctly with the unit vectors. Recall that rotation of a point CCW is the same as rotation of the coordinate system CW, so we use R_{cw}:

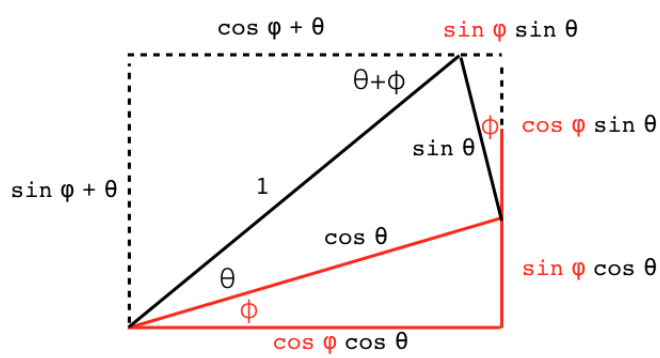
$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$

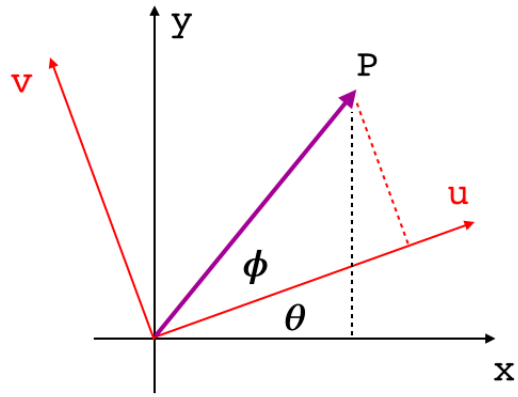
The rotated unit vector $\hat{\mathbf{j}}$ lies in the second quadrant.

Stewart

I found a nice method in Stewart which depends on knowing the sum of angles formulas.



It goes like this.



Here, we have the standard xy-coordinates in black. The rotated uv-coordinate system is in red, with a rotation angle θ in the counter-clockwise direction.

The ray to the point P has length r. The coordinates of point P in the u, v system are naturally expressed in terms of ϕ:

$$u = r\cos\phi$$

$$v = r\sin\phi$$

while x and y are naturally expressed in terms of the combined angle $\theta + \phi$.

$$x = r\cos(\theta + \phi)$$

$$y = r\sin(\theta + \phi)$$

Now, use the sum formula for cosine:

$$x = r\cos\theta\cos\phi - r\sin\theta\sin\phi$$

We have from above that

$$u = r\cos\phi$$

$$v = r\sin\phi$$

So

$$x = u\cos\theta - v\sin\theta$$

It's as easy as that.

In the same way:

$$\begin{aligned} y &= r\sin(\theta+\phi) \\ &= r\sin\theta\cos\phi + r\cos\theta\sin\phi \\ &= u\sin\theta + v\cos\theta \end{aligned}$$

Notice that here we have x, y in terms of u, v. So this derivation gives R_{cw}.

algebraic interconversion

Given $x = f(x, y)$ and $y = g(x, y)$, we wish to convert these to $u = f'(x, y)$ and $v = g'(x, y)$

One way is to switch symbols x, y for u, v, and at the same time, substitute $-\theta$ for θ. What this amounts to is relabeling our diagram with x, y being the rotated axes, and then rotating in the opposite direction (cw rather than ccw).

$$x = u\cos\theta - v\sin\theta$$

switch

$$\begin{aligned} u &= x\cos -\theta - y\sin -\theta \\ &= x\cos\theta + y\sin\theta \end{aligned}$$

(Recall that $\cos -x = \cos x$ and $\sin -x = -\sin x$).

For y

$$y = u\sin\theta + v\cos\theta$$

switch

$$v = x\sin -\theta + y\cos -\theta$$

$$v = -x\sin\theta + y\cos\theta$$

Straightforward computation is easy as well.

$$x = u\cos t - v\sin t$$
$$y = u\sin t + v\cos t$$

Multiply the first equation by $\cos t$ and the second by $\sin t$ and add:

$$x\cos t + y\sin t = u(\cos^2 t + \sin^2 t) - v\sin t\cos t + v\sin t\cos t$$
$$= u$$

Multiply the first equation by $-\sin t$ and the second by $\cos t$ and add

$$-x\sin t + y\cos t = v$$

vectors

Another way is to start from vectors and matrices and remember that we have a 2×2 matrix.

Suppose the angle is $\theta = 0$. Then we want the matrix to be the identity matrix

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

That naturally means that $\cos\theta$ is for entries along the diagonals and $\sin\theta$ at the other positions.

We also want to rotate the unit vector $\hat{\mathbf{i}}$, to become unit vector $\hat{\mathbf{j}}$.

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \qquad \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

That looks good. So

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$

means that

$$u = x\cos\theta - y\sin\theta$$
$$v = x\sin\theta + y\cos\theta$$

Rotating the points counter-clockwise is equivalent to rotating the coordinate system clockwise.

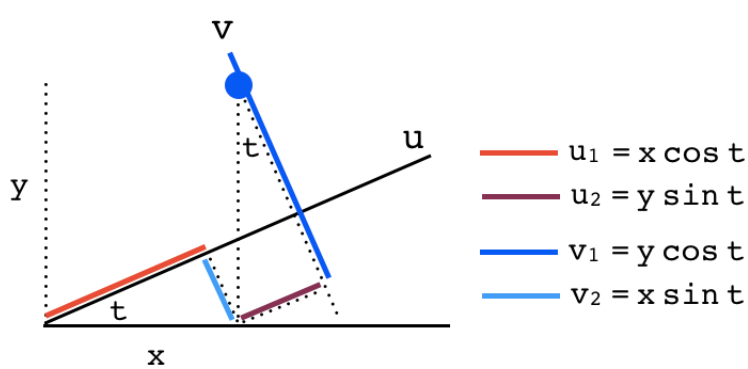
rotation by $+\theta$ then $-\theta$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

What follows is my original version of the first derivation.

standard derivation

Here is a second derivation with a sort of minimalist diagram of rotation



We draw the horizontal x-axis and the rotated u-axis. The angle between them is t. We plot our point and then draw perpendiculars to both axes. To finish the set-up, we draw perpendiculars from the point $(x, 0)$ as shown.

Once the drawing is rendered, we are almost done. Now we just work our way through.

$$u = u_1 + u_2 = x \cos t + y \sin t$$

All the triangles in the diagram are similar, with small angle t.

$$v = -v_2 + v_1 = -x \sin t + y \cos t$$

I find this hard to remember. You will know that you've done it right if you have both x and y as the *hypotenuse of a right triangle.* Start with the one for x and then rotate the triangle the other direction about the point $(x, 0)$. Stretch to fit.

Here we are rotating the coordinate system counter-clockwise. Hence, the minus sign is in the formula for the *vertical* component, v.

examples

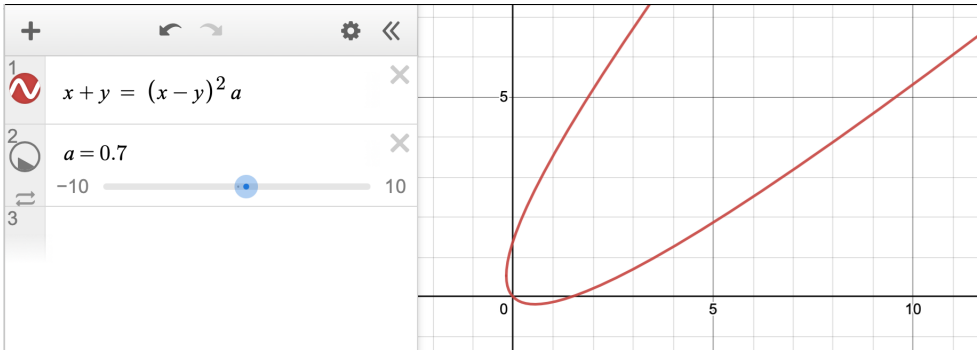
Consider $y = x^2$ and rotate through an angle of $45°$ cw. The points are rotated cw and the coordinate system ccw, so the minus sign is in the expression for x'.

$$x' = 1/\sqrt{2}(x - y)$$
$$y' = 1/\sqrt{2}(x + y)$$

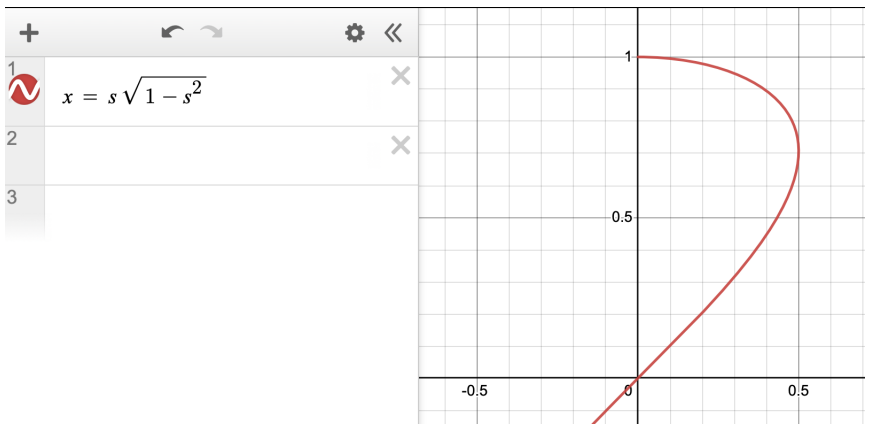
so

$$y' = x'^2$$
$$1/\sqrt{2} \cdot (x + y) = 1/2 \cdot (x - y)^2$$

$$x + y = 1/\sqrt{2} \cdot (x - y)^2$$



Here's another form — both rotated and translated.



Going back to our first result:

$$x' = x\cos\theta - y\sin\theta$$

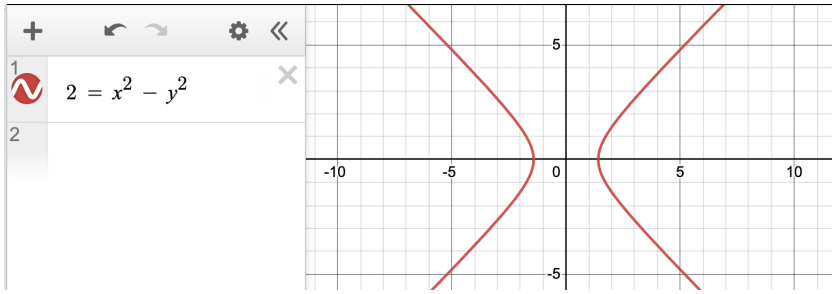
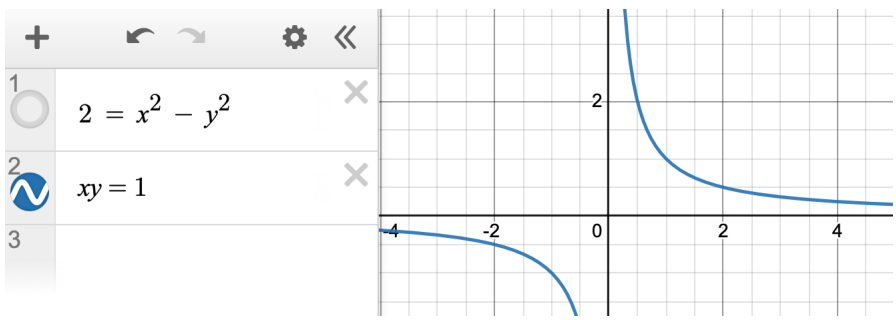
$$y' = x\sin\theta + y\cos\theta$$

Let $x'y' = 1$, that is, the graph is a hyperbola. We turn through an angle of 45° cw.

$$(x\cos\theta - y\sin\theta)(x\sin\theta + y\cos\theta)$$

$$\frac{1}{2} \cdot (x - y)(x + y)$$

$$2 = (x^2 - y^2)$$



When this guy is rotated by 45 ccw, it becomes the familiar graph of $xy = 1$.